National Aeronautics and
Space Administration

SLSTP FINAL PRESENTATION

Synthetic transcriptomics for mouse spaceflight

Can generated RNA-seq improve FLT vs GC analysis within each tissue?

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BPS

Biological & **P**hysical **S**ciences

QUESTION

Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

NASA OSDR

1,610 biological profiles

75 accessions

835 FLT

775 GC

ARCHS4 mouse

997,515 profiles audited

17,244 selected

20 tissues

- Mission, strain, material and assay differences can resemble a flight effect.
- The generator must preserve tissue and FLT/GC structure without memorizing samples.
- Statistical evidence must still come from observed OSDR profiles.

Sources: NASA OSDR Biological Data API; ARCHS4 mouse v2.5.

METHOD

We compared three model families and used conditional diffusion

The model architecture followed each paper; preprocessing, harmonization and conditioning remained configurable.

Expression

TPM
974 mouse landmarks
train-fitted MaxAbs

Models

WGAN-GP
DDIM
GeneJEPA screen

Training

ARCHS4 pretrain
OSDR adaptation
pooled tissue model

Conditions

tissue + FLT/GC
accession + material
matched profile generation

Selected branch

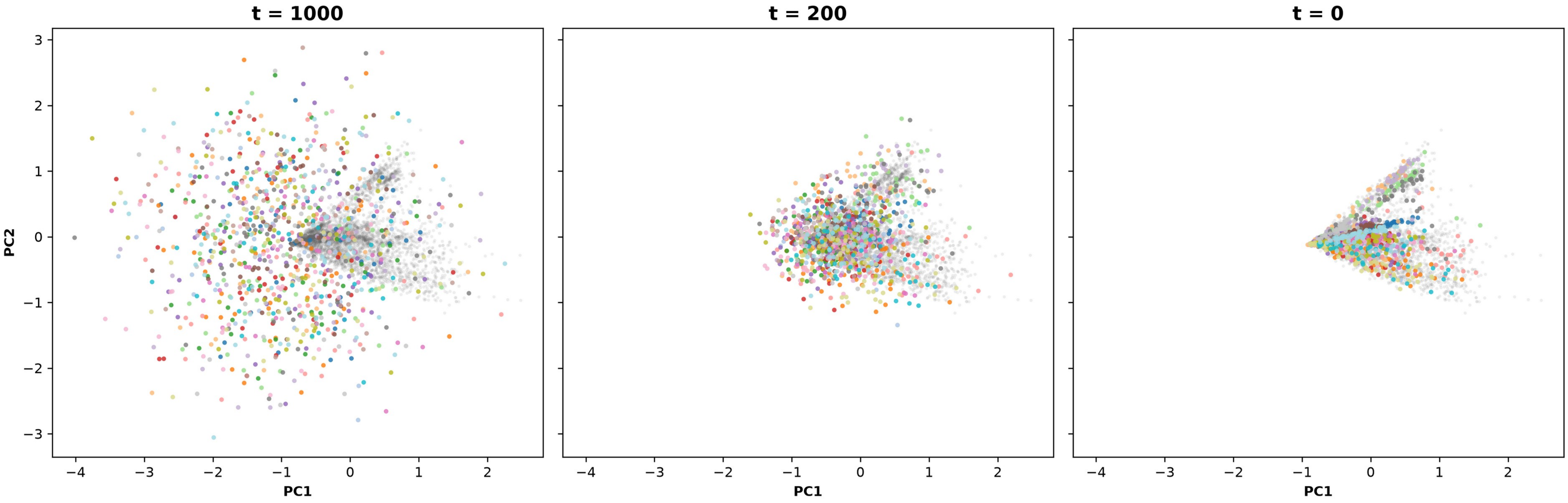
No global batch correction. Accession was supplied as a model condition so study structure was represented instead of erased.

Nine liver harmonization arms and a 463-row experiment plan were screened before full training.

Model sources: Vinas et al., Bioinformatics (2022); Lacan et al., BMC Bioinformatics (2026); GeneJEPA (2025).

Diffusion learns tissue structure from noise

The same PCA axes follow 1,024 generated profiles through reverse diffusion.



t = 1000: noise

t = 200: partial structure

t = 0: tissue-conditioned profiles

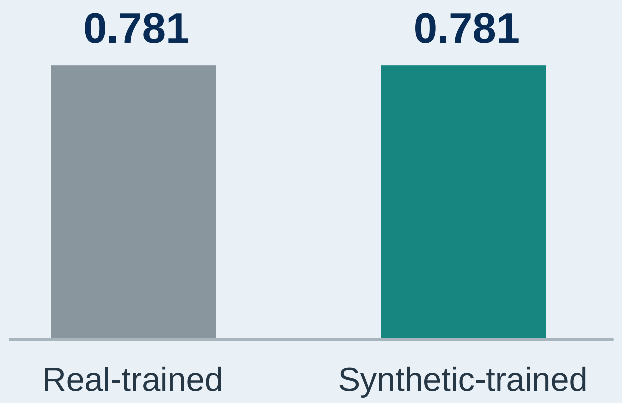
Gray points are real ARCHS4 profiles; colors identify generated tissue conditions. PC1 and PC2 limits are shared across panels.

DDIM matched expression and reduced separability

AA closer to 0.5 means a classifier has less success separating real and generated profiles; lower FD is better.

ARCHS4 tissue probe

Balanced accuracy on held-out real profiles



Metric	WGAN-GP	DDIM	Reading
Correlation	0.976	0.974	similar
F1	0.985	0.997	higher
Adversarial accuracy	0.636	0.475	closer to 0.5
FD / real P95	0.144	0.074	lower

Use DDIM downstream

Lower separability and distributional distance, with comparable correlation and higher F1.

WGAN values use validation data and DDIM values use the stated OSDR test; they are not paired on one common split.

ANALYSIS

Synthetic profiles entered the analysis in five different ways

Every arm was judged on held-out real profiles before synthetic attribution was allowed.

Real OSDR
observed FLT / GC

DDIM generated
matched conditions

Real only
rank: real
fit: real

Generated only
rank: generated
fit: generated

Real + generated
rank: consensus
fit: equal weight

Guided, real fit
rank: consensus
fit: real

Guided, 5% synth.
rank: consensus
fit: real + 5%

Held-out real profiles

BA | AUROC | AP

select one eligible arm
for each tissue

Biology check FLT vs GC effects, random-effects meta-analysis and BH FDR use real OSDR profiles only.

Synthetic profiles never increase animal n.

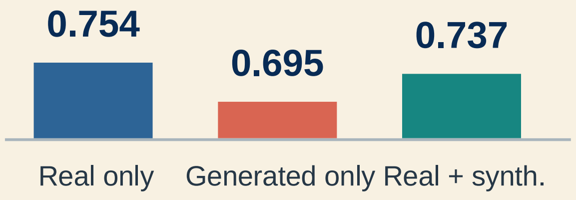
UTILITY

Pooling tissues hid useful signal

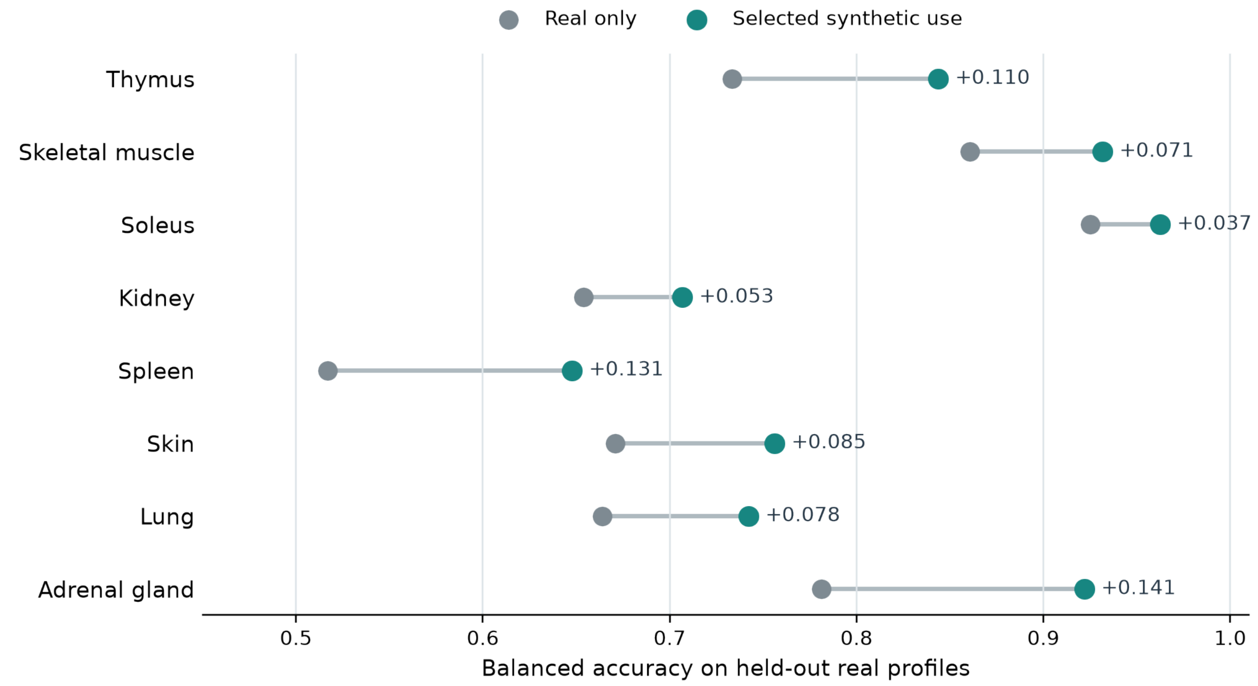
A single FLT/GC classifier did not improve, but tissue-specific synthetic use often did.

Pooled across tissues

Balanced accuracy



Selected use within each tissue



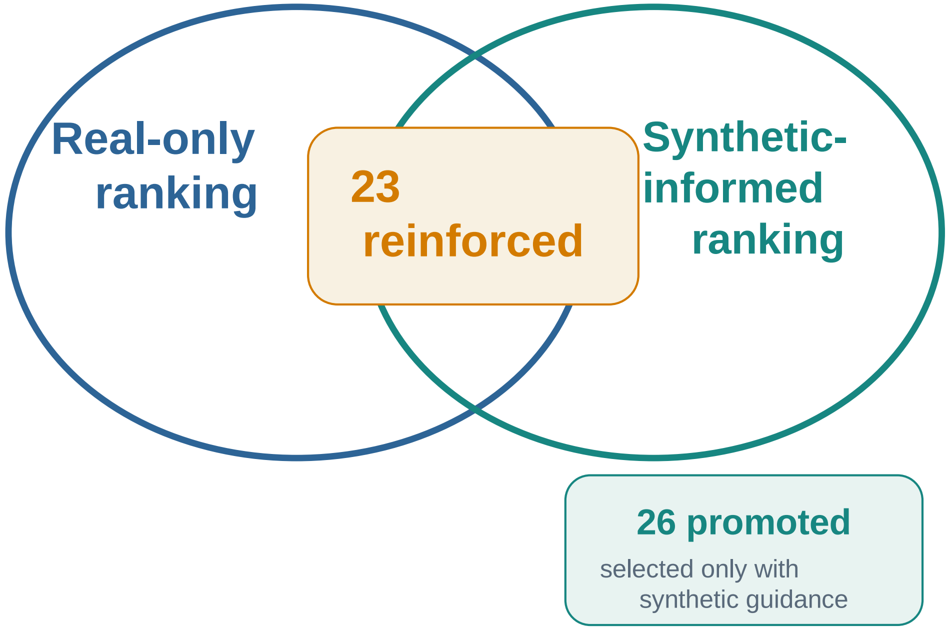
The useful arm differed by tissue.

FEATURE ANALYSIS

Synthetic guidance changed ranking, not statistical evidence

Promoted and reinforced describe repeated feature selection; both require a supporting effect in real OSDR data.

Repeated stable selection



Real OSDR association test

- Estimate FLT vs GC inside each accession
- Combine accession effects with a random-effects model
- Apply BH FDR within each tissue

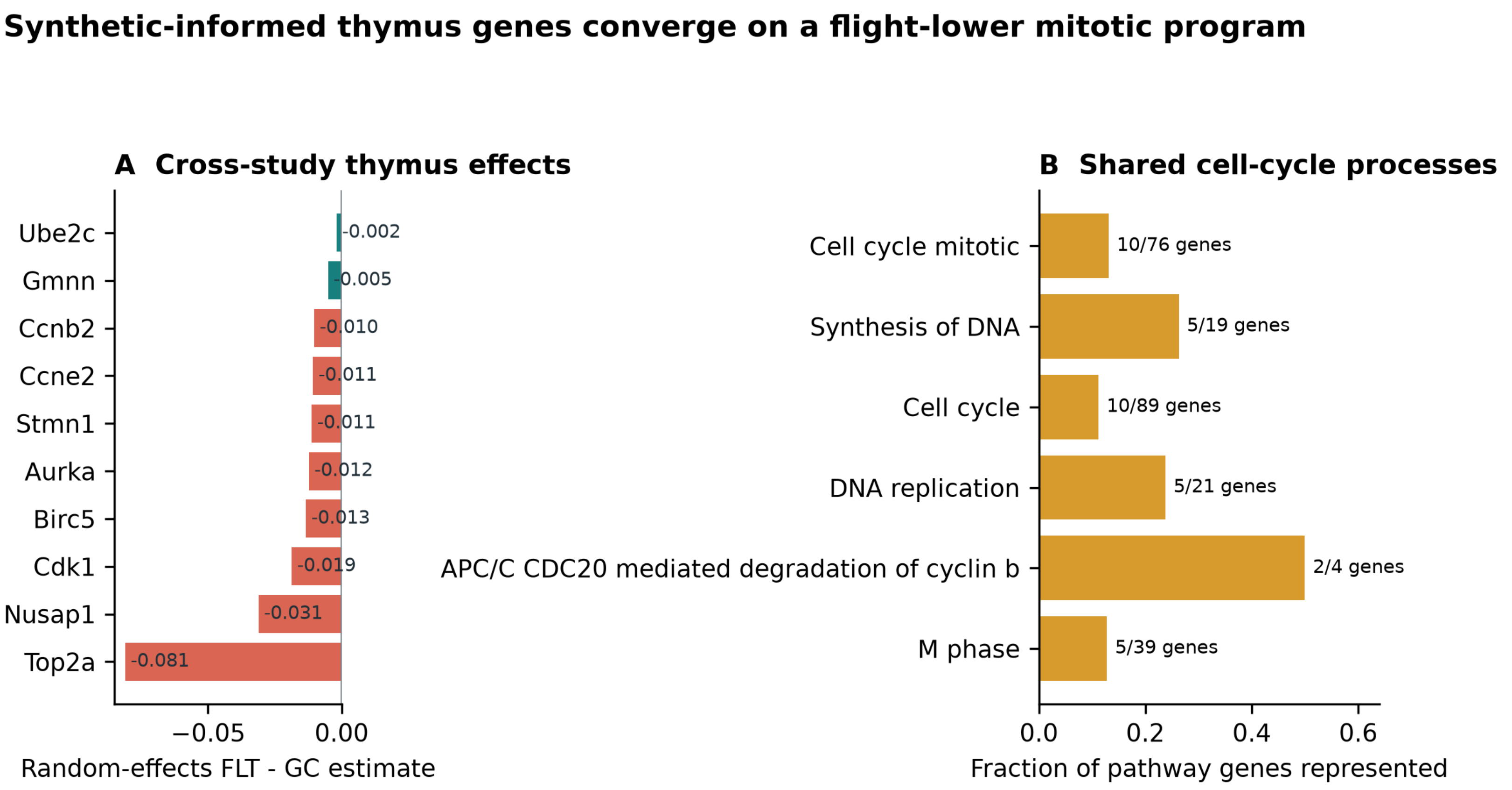
49 synthetic-informed associations

These are 49 tissue-gene associations among 459 BH-FDR rows. A gene can appear in more than one tissue.

THYMUS

Thymus points to lower proliferative renewal

Synthetic guidance organized real-data associations into a coherent mitotic and DNA-replication panel.



16 synthetic-informed BH-FDR associations

13 promoted | 3 reinforced

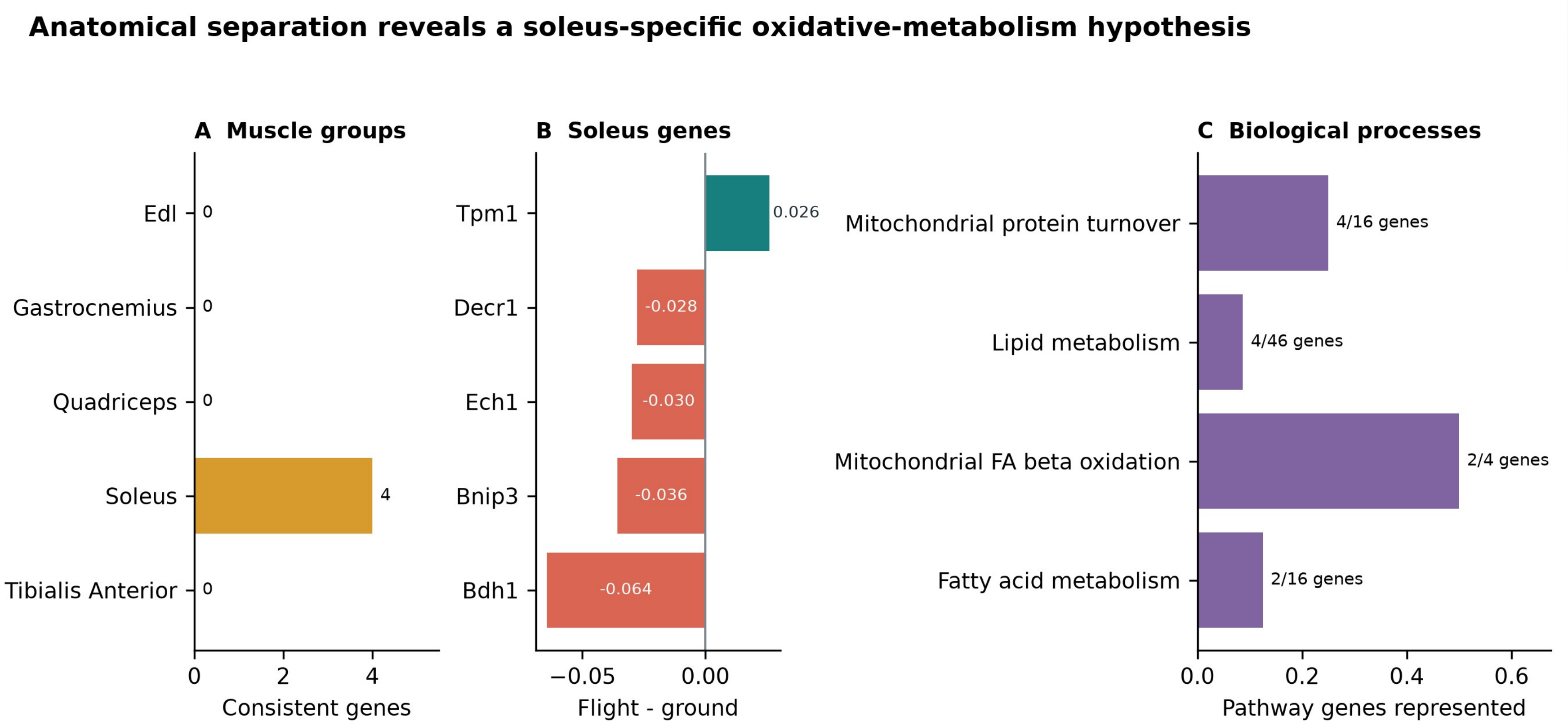
- Lower Nusap1, Stmn1, Birc5, Cdk1, Top2a and related genes
- Cell cycle, DNA replication, APC/C and G2/M processes
- Bulk RNA-seq cannot separate cell loss from lower transcription

Prior thymus studies report involution and altered cell-cycle expression after spaceflight (Gridley et al. 2013; Horie et al. 2019).

SOLEUS

Soleus reinforces a mitochondrial and lipid program

This result strengthens an existing real-data pattern rather than promoting a new soleus gene.



0.925 -> 0.963

balanced accuracy

5 reinforced | 0 promoted

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Compatible with unloading and contractile remodeling

Literature context: Gambara et al. (2017) and Stein et al. (2002).

Kidney adds a focused signal; other tissues are exploratory

The strongest process-level stories were thymus and soleus, but several tissues produced narrower candidates.

Kidney	Inpp4b promoted, Slc37a4 reinforced; both higher in flight	**Focused secondary**
Spleen	Rai14, Ptprk and Myl9 promoted; Loxl1 reinforced; no Reactome enrichment	**Exploratory**
Skin + adrenal	Plscr1 higher in skin; Psmb8 and Tspan4 lower in adrenal	**Exploratory**
Lung + retina	Predictive gains without a synthetic-informed BH-FDR gene	**Prediction only**
Liver + EDL + quadriceps	Real-only arm retained	**No synthetic claim**

Independent missions are still needed before any candidate becomes a transferable spaceflight biomarker.

TAKEAWAYS

Synthetic data helped most as a tissue-specific prior

The generator created useful structure, not new biological replication.

1 Generate

Conditional DDIM produced high-fidelity profiles with near-chance real-versus-synthetic separation.

2 Use

Pooled augmentation failed. Tissue-specific ranking and low-weight training were more useful.

3 Interpret

Thymus and soleus formed the clearest programs; kidney supplied a focused secondary pair.

Next test

Hold out complete accessions, then validate the thymus panel, soleus metabolism and kidney Inpp4b / Slc37a4 in independent data.

Generated profiles never enter the biological sample count or the BH-FDR test.

Thank you

Questions?

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Code and manuscript	github.com/jasont314/nasa-mouse

All biological associations were tested in observed OSDR profiles.